

Sentiment Polarity Classification Model Evaluation

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Model Performance Metrics

Metric	Value
True Positives (TP)	664
True Negatives (TN)	679
False Positives (FP)	167
False Negatives (FN)	152
Precision	0.81
Recall	0.80
F1-Score	0.81

Table: Model evaluation metrics

Approach

- ▶ Data Preprocessing:
 - ▶ Loaded positive and negative sentiment data
 - ▶ Labeled and shuffled the dataset
 - ▶ Split into training(8000), validation(1000), and test sets(1662)
- ▶ ELMo Embedding:
 - ▶ Converted text data to ELMo embeddings
- ▶ Classification:
 - ▶ Used Logistic Regression as the classifier
 - ▶ Achieved validation accuracy of 79.1%
 - ▶ Test accuracy of 80.75%

Logistical Information

- ▶ Software Required:
 - ▶ Python 3.x
 - ▶ Jupyter Notebook / Google Colab
- ▶ Packages Needed:
 - ▶ pandas
 - ▶ numpy
 - ▶ scikit-learn
 - ▶ tensorflow
 - ▶ tensorflow_hub (for ELMo)
 - ▶ nltk (for data processing)
- ▶ Link to Code:
 - ▶ [link](#)

Conclusion

- ▶ The model achieved good accuracy on the sentiment analysis task.
- ▶ Future improvements could involve using more complex models or fine-tuning hyperparameters.

References

- ▶ Original dataset:
<https://www.cs.cornell.edu/people/pabo/movie-review-dat>
- ▶ ELMo: Peters et al. (2018). *Deep Contextualized Word Representations*.